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1  # Full RF Rectifier Report
2
3  ## 1. Optimization Simulation Setup
4  | Item | Value |
5  |-----|-----|
6  | RF Frequency (Hz) | 920 MHz |
7  | RF Input Amplitude (V) | 0.5 |
8  | Temperature (°C) | 27 |
9  | Supply Voltage VDD (V) | 0 |
10 | Process Corner | tt |
11 | Population Size | 50 |
12 | Max Generations | 30 |
13 | Fscale | 0.7 |
14 | Crossover Rate (CR) | 0.8 |
15 | Stall Generation Limit | 10 |
16 | Acceptance Tolerance | 1e-12 |
17 | Stop Cost Threshold | acceptTol |
18 | PVT Corners | tt, ss, ff, sf, fs |
19 | VDD Sweep (V) | 0 |
20 | Temperature Sweep (°C) | -40, 0, 27, 70, 85 |
21
22 ---
23
24 ## 2. Best Solution Summary
25 | Parameter | Value |
26 |-----|-----|
27 | Generation | 12 |
28 | Index | 37 |
29 | Total Index | 687 |
30 | VOUTMAX | 0.503 V |
31 | POUT | 1.21 µW |
32 | PIN | 1.30 µW |
33 | PCE | 92.72 % |
34 | Sensitivity (PIDBM) | -28.86 dBm |
35 | Ripple (ΔV) | 5.21 mV |
36 | Area | 641 µm2 |
37 | FOM1 | 665.7 |
38 | FOM2 | 19211.3 |
39 | FOM3 | 38171.1 |
40
41 ---
42
43 ## 3. Constraint Verification
44 | Metric | Target | Result | Status |
45 |-----|-----|-----|-----|
46 | VOUT | 0.5-1 V | 0.503 V | PASS |
47 | POUT | ≥1 µW | 1.21 µW | PASS |
48 | PCE | 85-95 % | 92.72 % | PASS |
49 | PIDBM | -35 to -15 dBm | -28.86 | PASS |
50 | Ripple | ≤10 mV | 5.21 mV | PASS |
51 | Area | within bounds | 641 µm2 | PASS |
52
53 ---
54
55 ## 4. PVT Summary (Extremes)
56 - VOUT: **0.472 - 0.524 V**
57 - POUT: **1.06 - 1.31 µW**
58 - PCE: **70.6 - 95.7 %**
59 - PIDBM: **-29.06 to -28.10 dBm**
60 - Ripple: **4.18 - 8.17 mV**
61
62 ---
63
64 ## 5. Monte Carlo Summary
65 | Metric | Mean | Sigma |
66 |-----|-----|-----|
67 | VOUT | 0.503 V | 2.55 mV |
68 | POUT | 1.205 µW | 11.9 nW |
69 | PCE | 92.45 % | 1.85 % |

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70 | PIDBM | -28.63 dBm | 0.24 dB |
71 | Ripple | 5.17 mV | 0.196 mV |
72
73 ---
74
75 ## 6. Yield Report
76 - Total Samples: 1000
77 - Passing Samples: 577
78 - **Yield: 57.70%**
79
80 ---
81
82 ## 7. RF Performance Metrics
83 - Dynamic Range (DR): **22.11 dB**
84 - PCE Peak: **91.88 % @ -29.55 dBm**
85 - Sensitivity (@1 V): **-16.97 dBm**
86
87 ---
88
89 ## 8. Impedance Matching (X-Ray)
90 - Input Impedance (pre-match): **1.01 - j148 Ω**
91 - S11 (before): **-0.036 dB**
92 - S11 (after): **-23.89 dB**
93 - Matched Impedance: **≈ 50 Ω**
94
95 ---
96
97 ## 9. Runtime Summary
98 - Total Runtime: **2.68 hours**
99
```